

SUMMARY

Software Engineer with hands-on experience in **backend development and AI/ML** through defence and industry internships and technical projects. Experienced in developing **Java and Spring Boot applications, RESTful services, and data-driven solutions**, with additional exposure to **computer vision, NLP, and databases**. Demonstrates strong **problem-solving, ownership, and adaptability**, with a focus on building clean, maintainable software and translating requirements into practical solutions.

EXPERIENCE

Indian Army Internship Program (AARIC)

Full Stack AI/ML Engineer

Jan 2026 – Mar 2026

Bengaluru, India

- Developed and validated defence-oriented software solutions, integrating **AI/ML pipelines, RESTful services, and secure cloud infrastructure** for internal applications.
- Contributed to **backend components and system design**, applying object-oriented software engineering principles to integrate AI/ML functionality with REST-based services.
- Developed and delivered an internal **web application for the Army Artificial Intelligence Research & Incubation Center (AARIC)**, providing a dedicated interface for the center's internal web presence: AARIC webpage.

SECENAI Semiconductors and Test Solutions Pvt. Ltd.

UAV Computer Vision & Data Analytics Intern

May 2025 – Jul 2025

- Worked on **UAV-based computer vision and data analytics** solutions during a **3-month internship**, processing aerial and sensor data to derive actionable insights.
- Applied **computer vision techniques and sensor-data analysis** to support drone-based analysis and data-driven decision-making workflows.

PROJECTS

Expense Splitter & Settlement Optimizer

Graph algorithms, Java, Spring Boot

- Designed a debt-simplification engine using **graph algorithms** to minimize transaction count in shared group expenses.
- Built **REST APIs** in **Java** and **Spring Boot** using controller/service/repository layering.

Real-Time Object Classification and Tracking (YOLOv8 & DeepSORT)

Computer Vision, Python, ML

- Built a **GPU-accelerated multi-object detection and tracking system** for classification and real-time video overlays.
- Achieved **75.95% precision, 60.67% recall, 61.08% F1-score, 69.96% mAP@0.5, and 53.23% mAP@0.5:0.95**.

Real-Time Sentiment Analysis on Twitter Data

MongoDB, VADER, Python

- Designed a **Python pipeline** to ingest, classify, and store **2.8M+ live tweets** with emoji-aware sentiment analysis.
- Achieved **85% classification accuracy** on a 1,000-tweet sample.

SKILLS

- Languages:** Java, Python, C, C++, SQL
- Frameworks / Tools:** Spring Boot, FastAPI, Streamlit, TensorFlow, PyTorch, Scikit-learn, YOLOv5–v8, DeepSORT, VADER, Git, Jira, n8n, Copilot, Docker
- Databases / Platforms:** MySQL, PostgreSQL, MongoDB, Linux, Power BI, MS Excel
- Concepts:** REST APIs, OOPs, Collections Framework, Multithreading, Generative AI, RAG, LLM Fine-tuning, Prompt Engineering, Computer Vision, NLP, Automation, Process Optimization

PUBLICATIONS AND PATENTS

A System and a Method for Monitoring Safety Measures During Workouts

Patent Id: 202441023005

Granted on: 12/01/2026

An Attendance Marking System and a Method Thereof

Patent Id: 202341067956

Publication: 02/10/2023

Comparative Paper on Sentiment Analysis using Machine Learning

Large-scale text classification on 160K+ reviews

Publication: 01/2023

EDUCATION

SRM University AP, Amaravati, India

Master of Technology, Data Science

Sep 2024 – Jun 2026

GPA: 8.71

SRM University AP, Amaravati, India

B.Tech in Computer Science Engineering (AI & ML)

Sep 2020 – Jun 2024

GPA: 8.43

CERTIFICATIONS

- HackerRank Certified: Java, Python, SQL, Problem Solving
- Oracle Database@AWS Architect Professional
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
- Oracle Data Platform 2025 Certified Foundations Associate

ACHIEVEMENTS

- Awarded 100% Merit Scholarship during M.Tech at SRM University AP.
- Gold Medal, Artificial Intelligence — SRM University Research Day 2022.
- Silver Medal, Image Processing and Computer Vision — SRM University Research Day 2023.